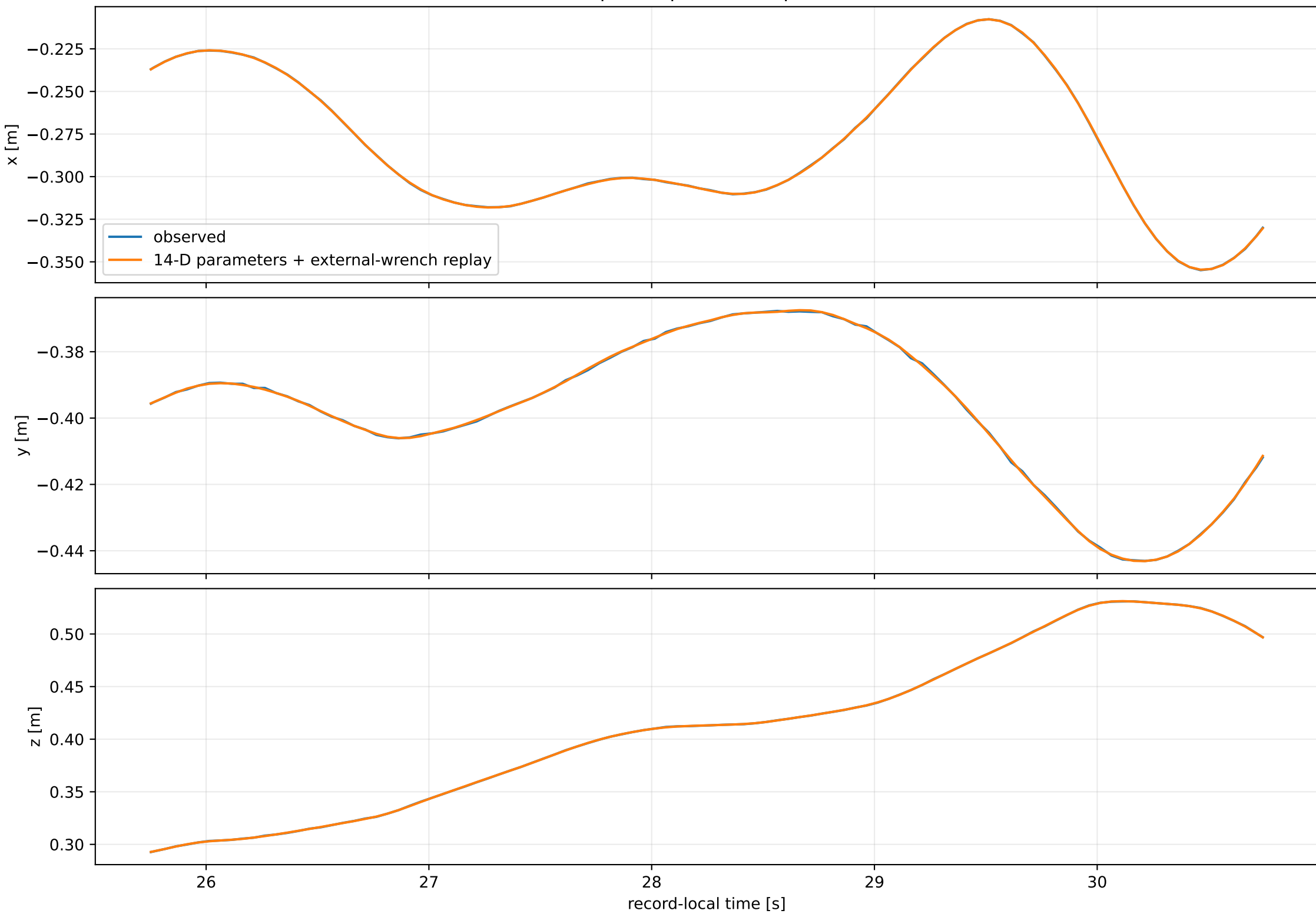
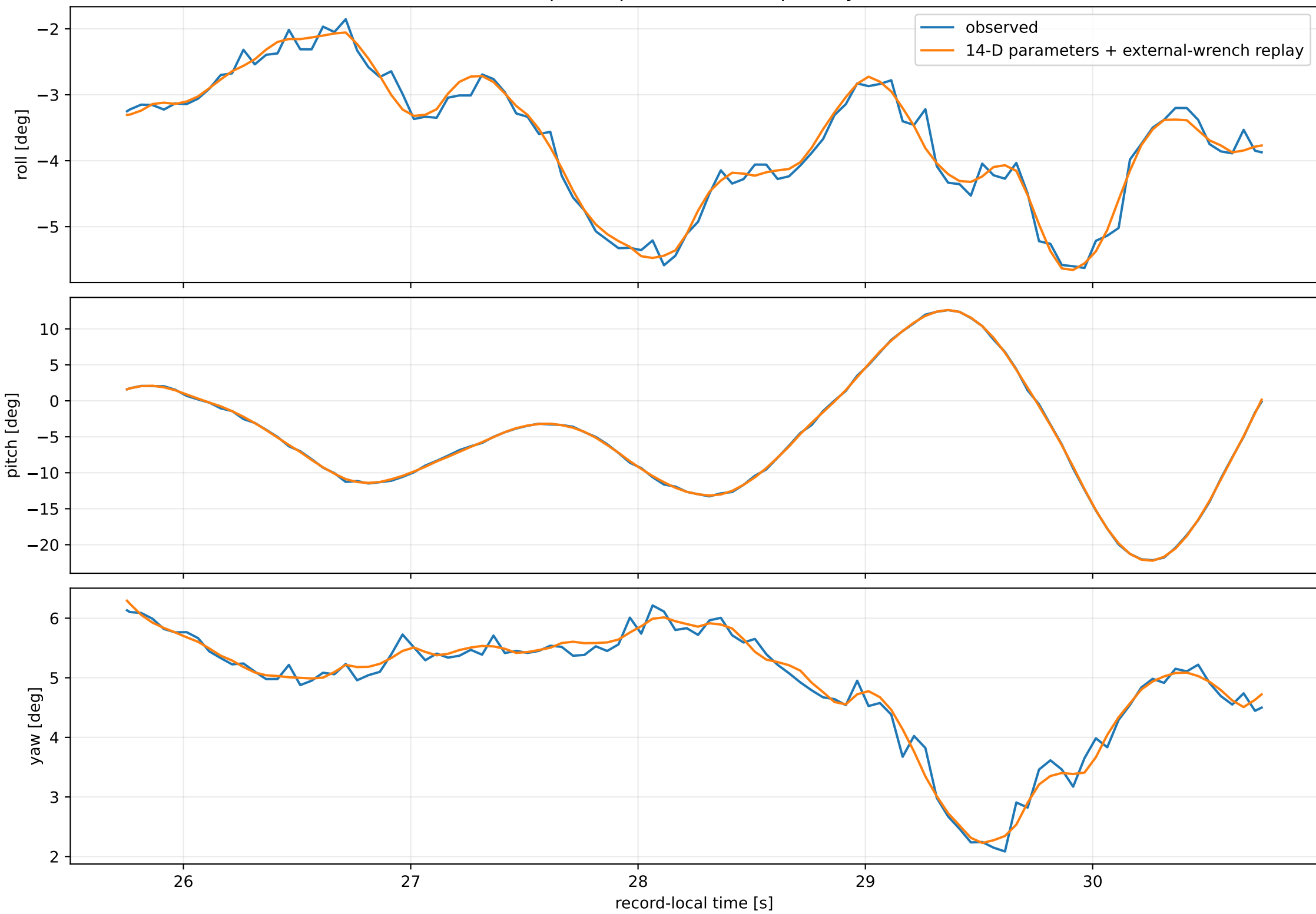


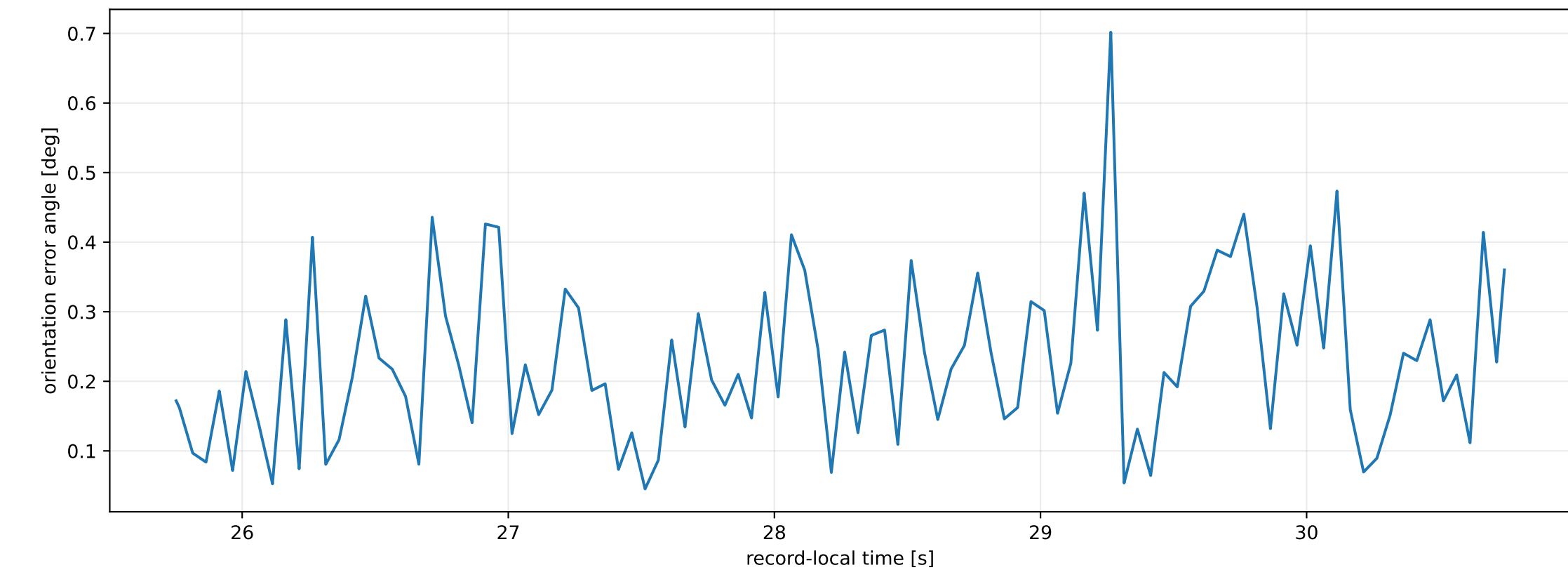
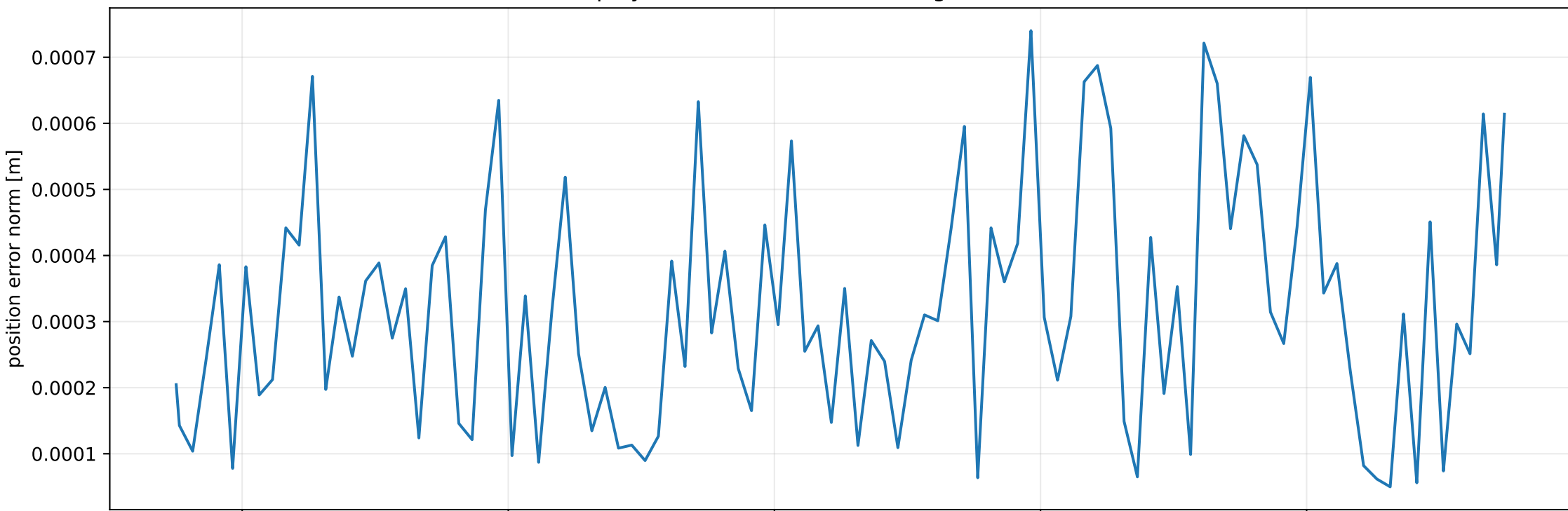
Fine pose reproduction: position



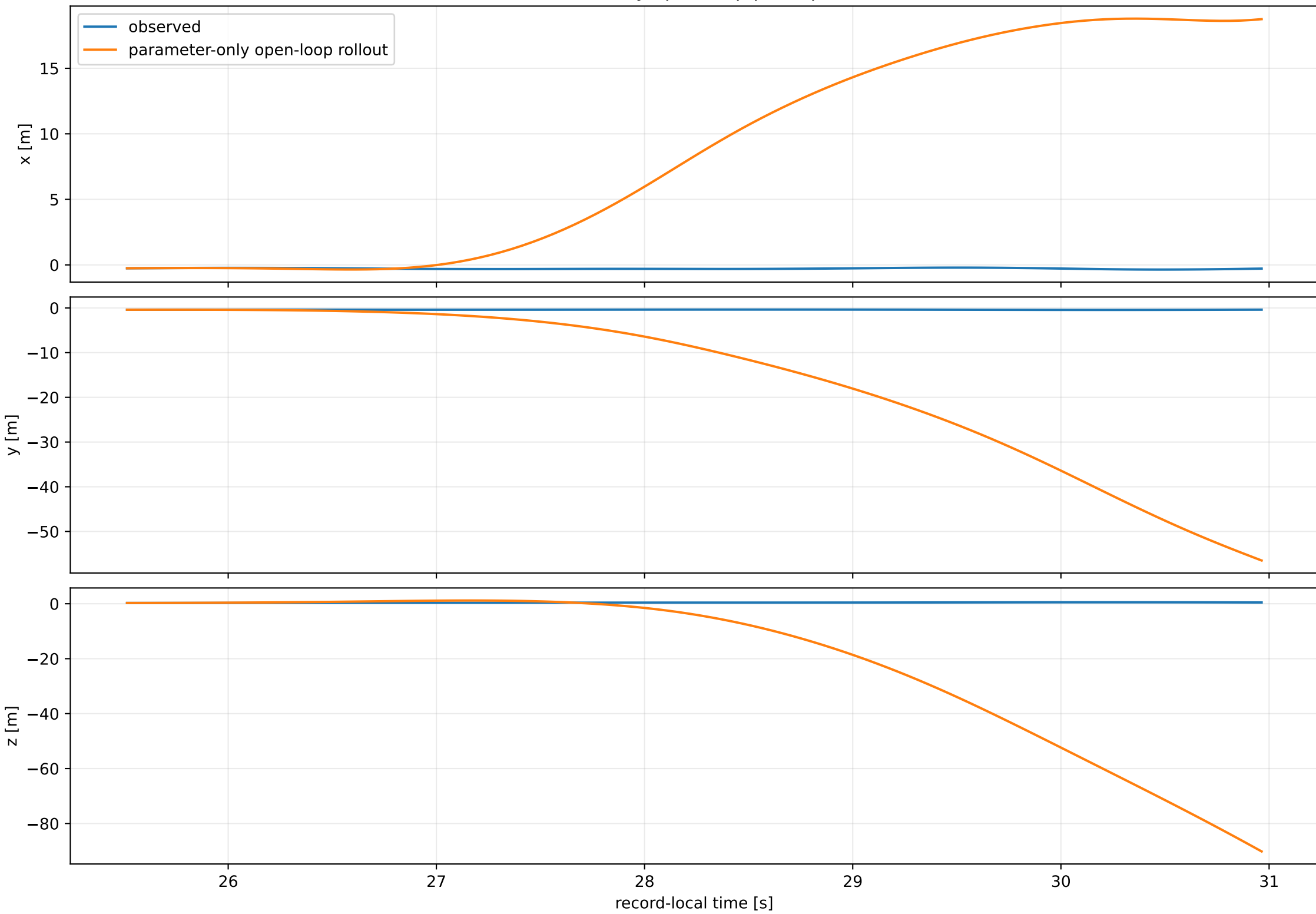
Fine pose reproduction: roll / pitch / yaw



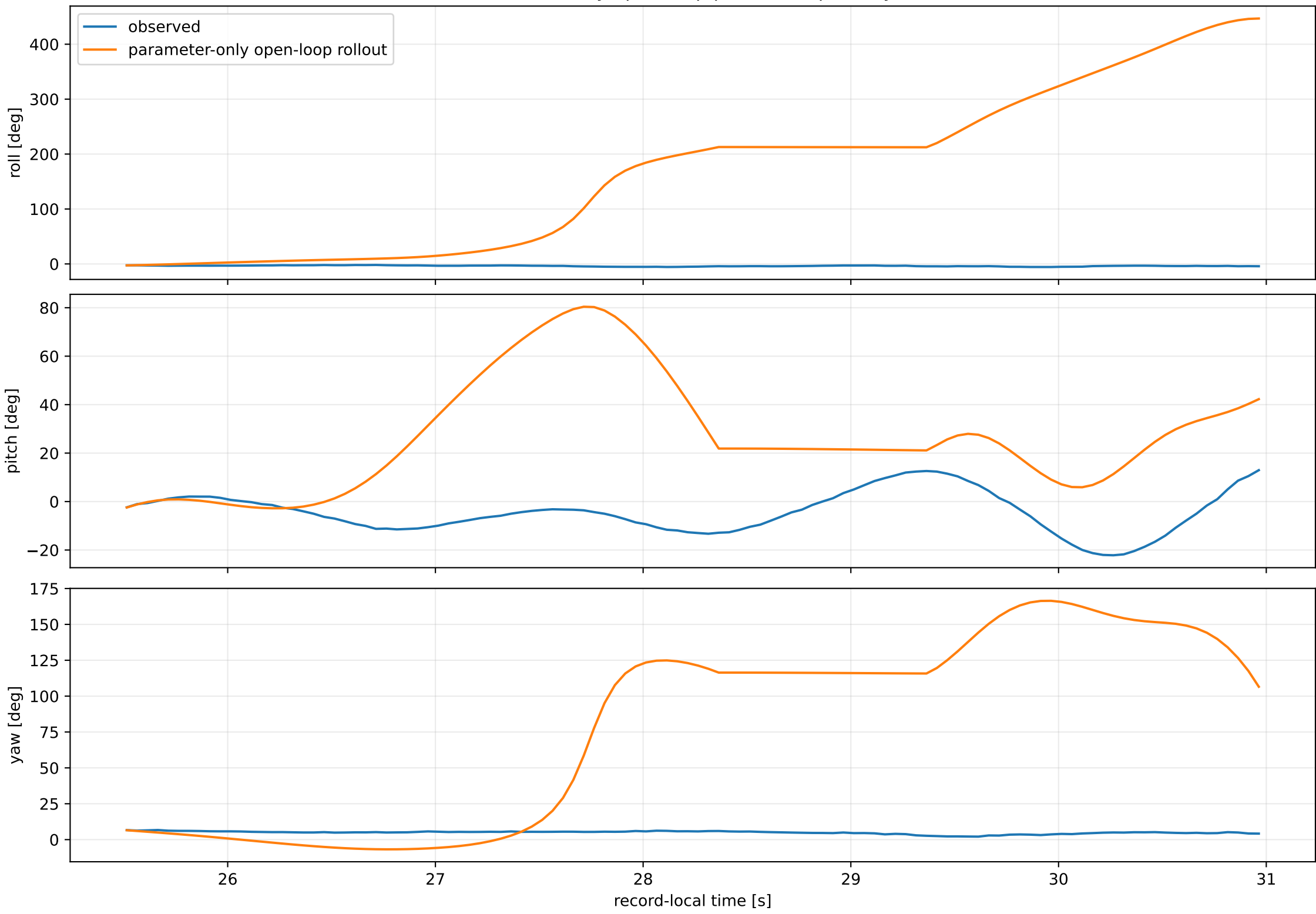
Fine replay error after the 14-D change — lower is better



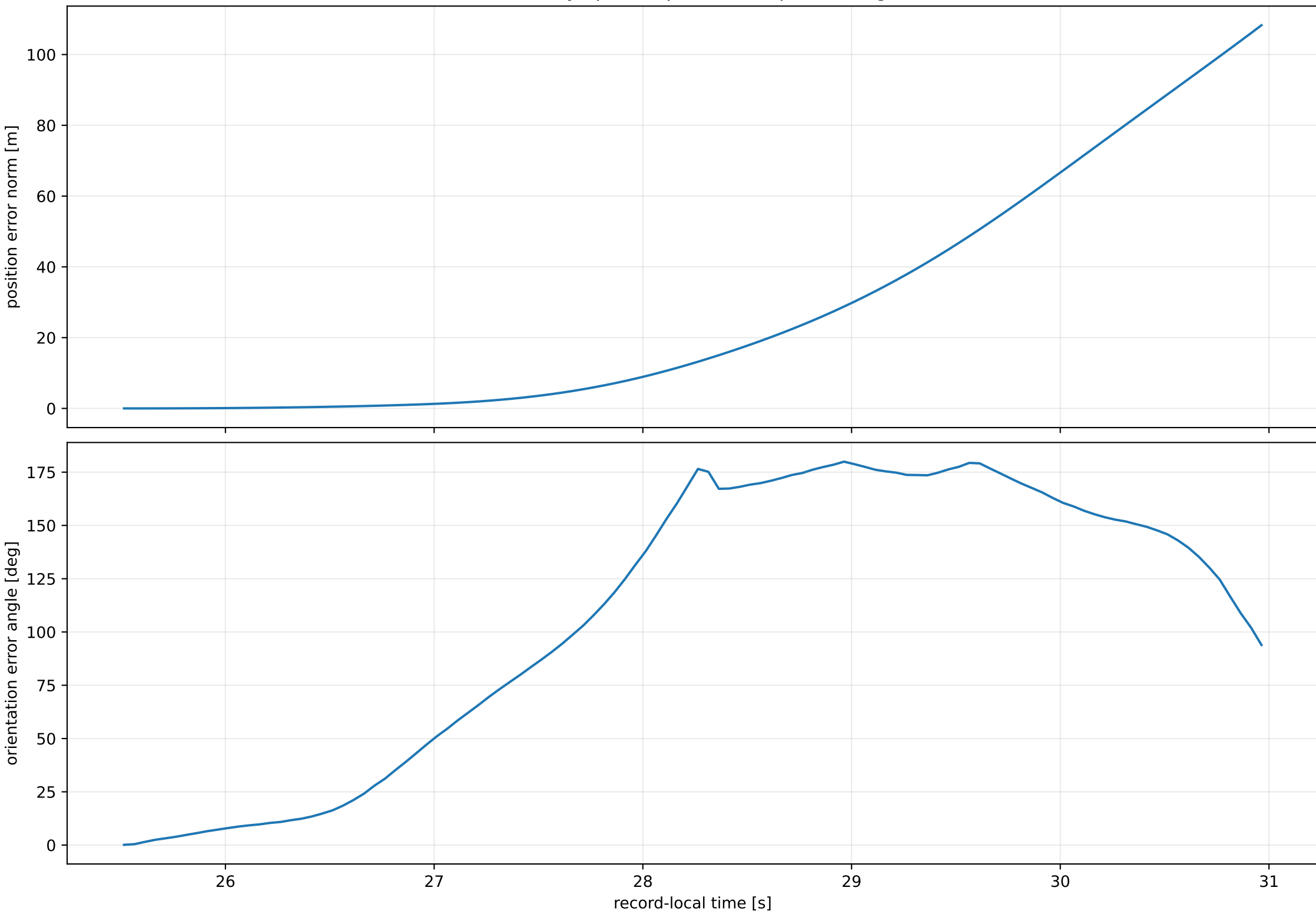
Parameter-only open-loop pose: position



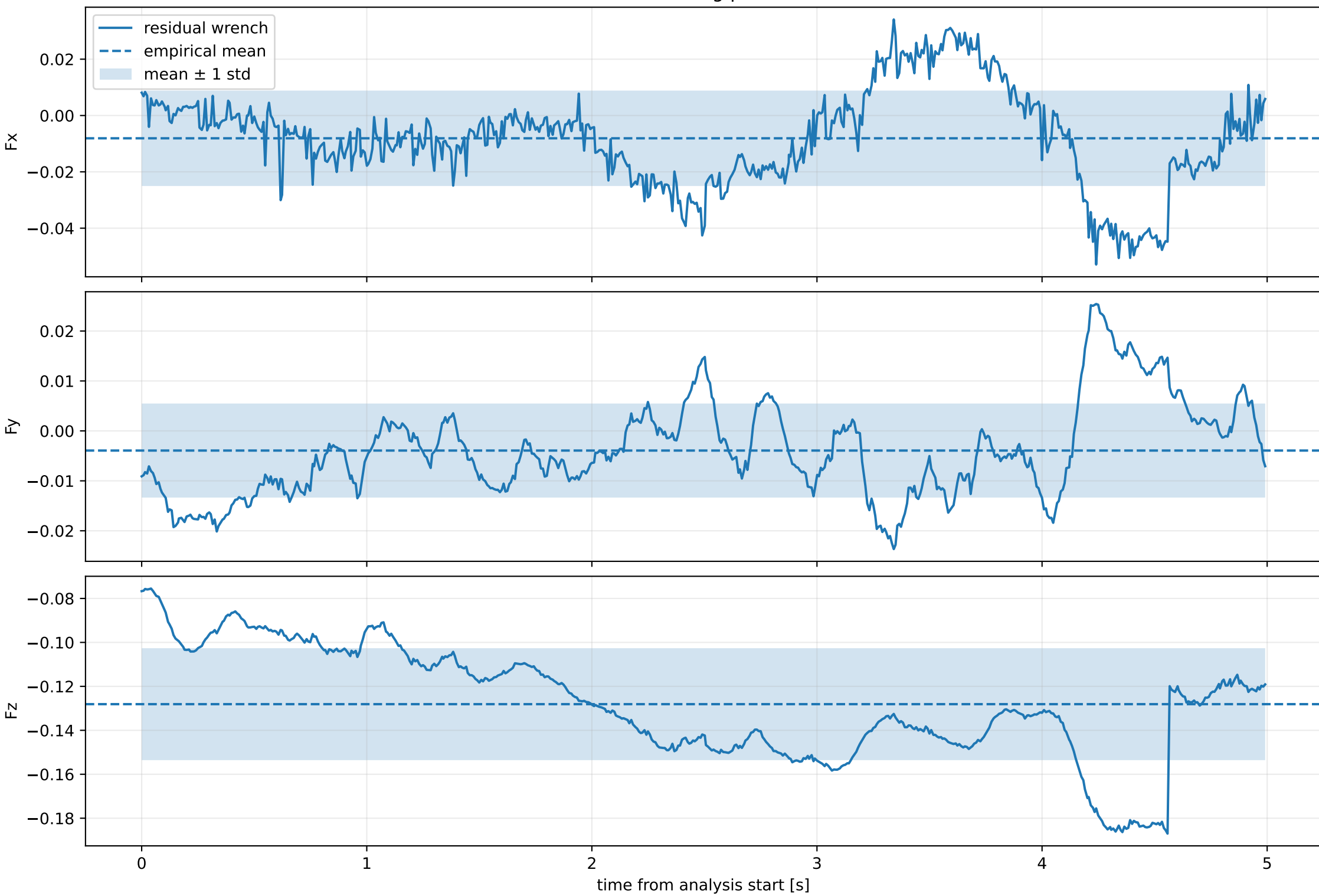
Parameter-only open-loop pose: roll / pitch / yaw



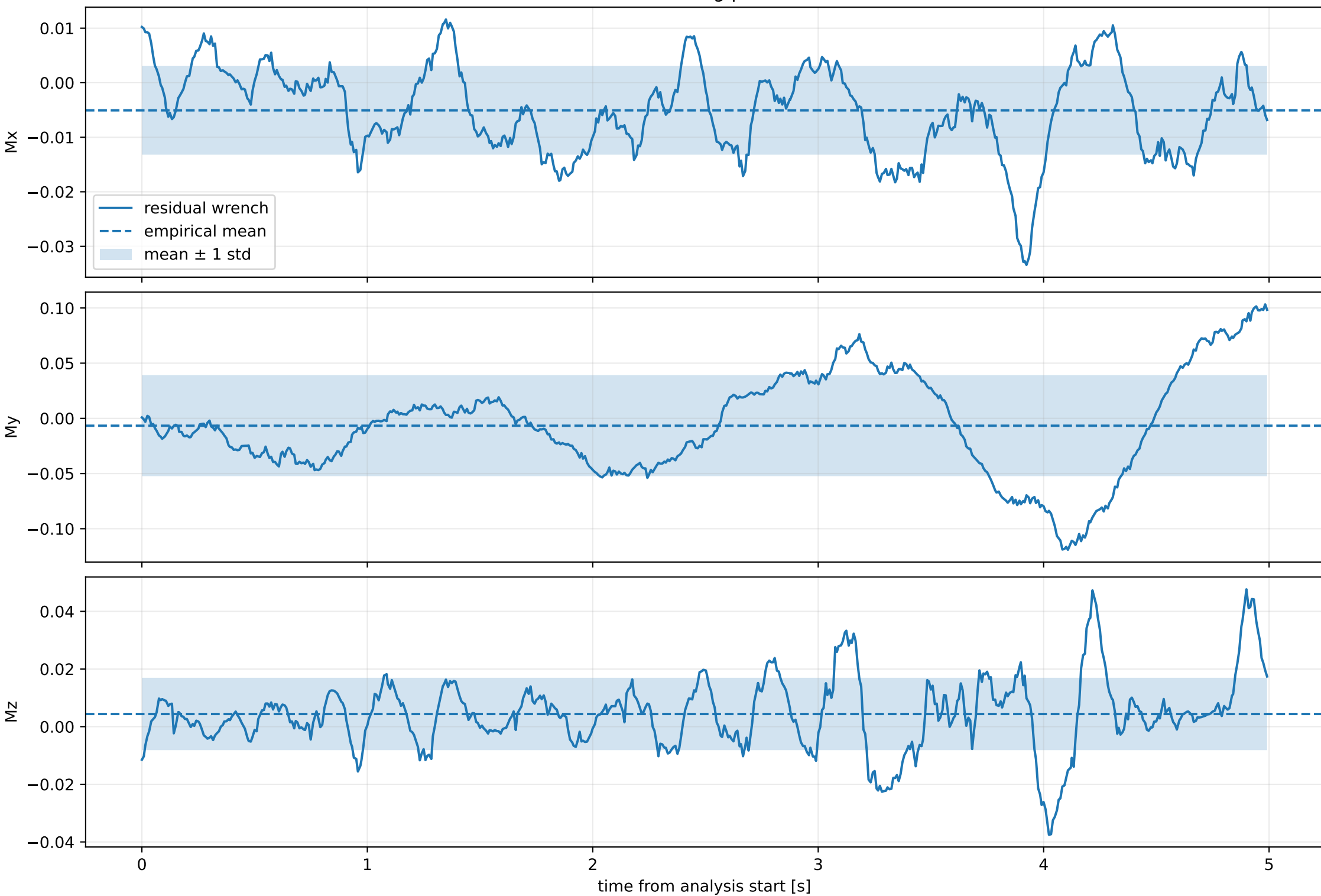
Parameter-only open-loop rollout (separate diagnostic)



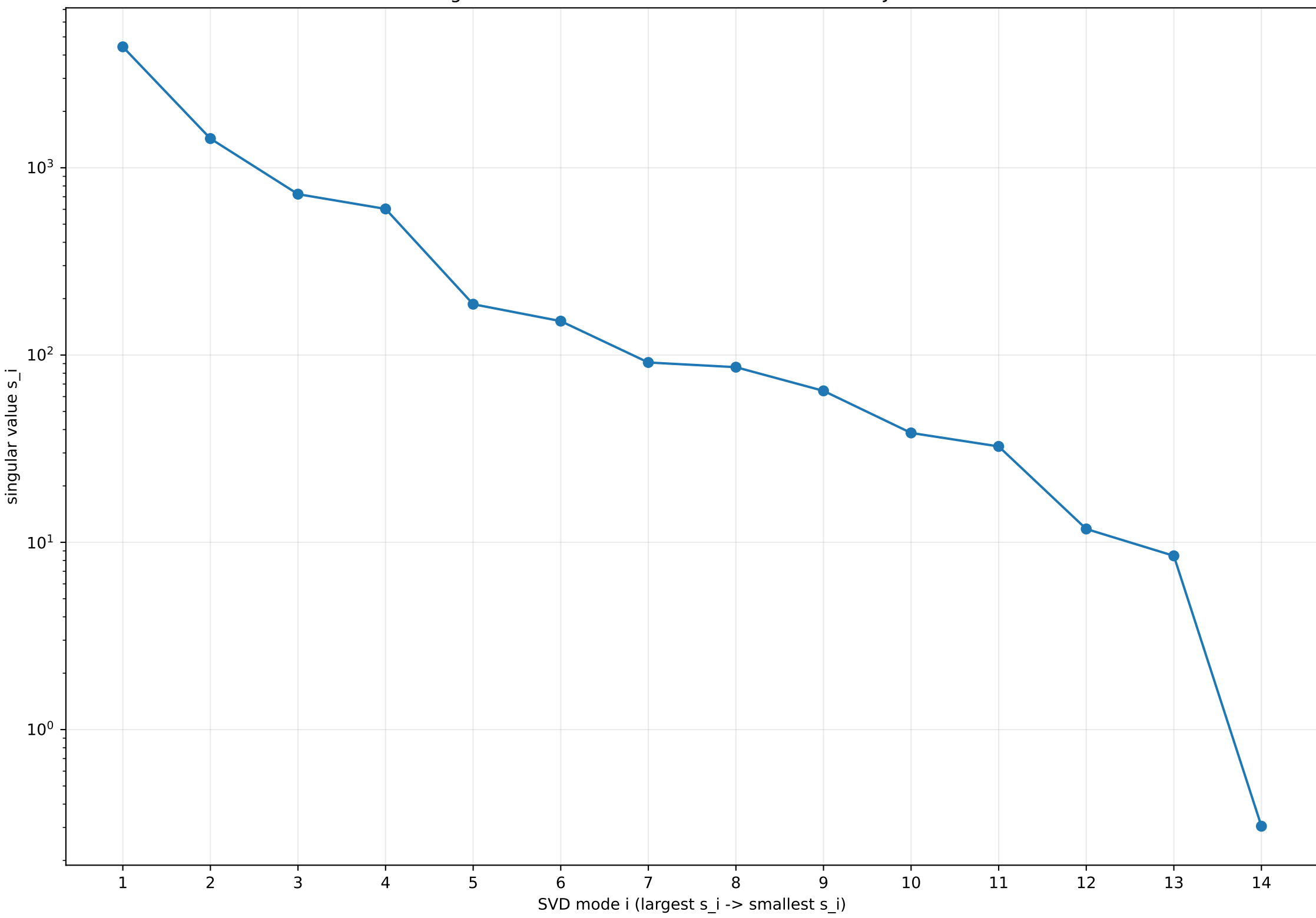
Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



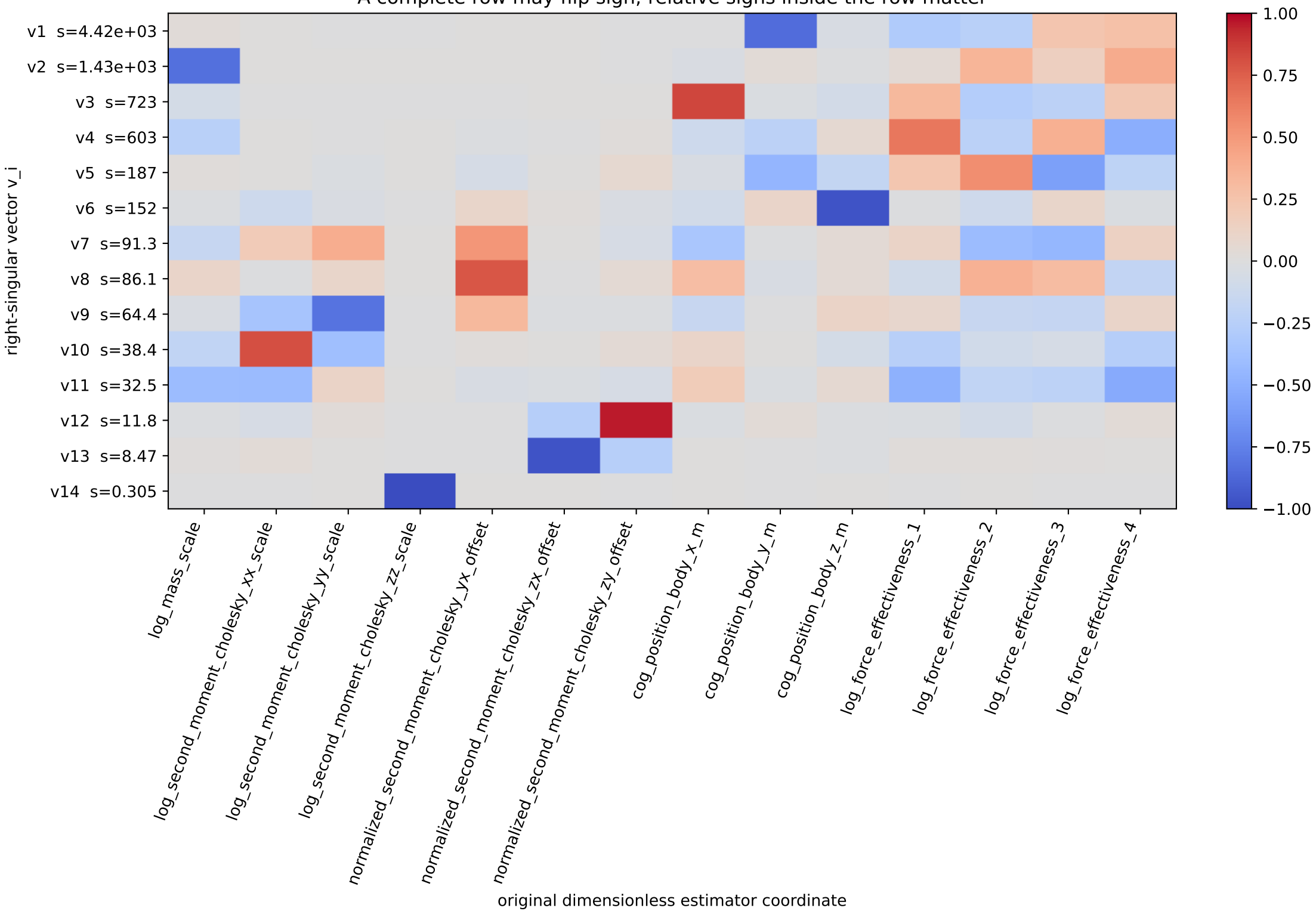
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.366227971

inertia J [kg m²]

[0.065002073 6.4295692e-06 1.8330468e-05]
[6.4295692e-06 0.065045086 3.15883e-06]
[1.8330468e-05 3.15883e-06 0.12899232]

principal moments [kg m²]

[0.065001127, 0.065046026, 0.12899233]

CoG position in body frame [m]

[-0.0025631903, 0.00037050893, 0.0094958309]

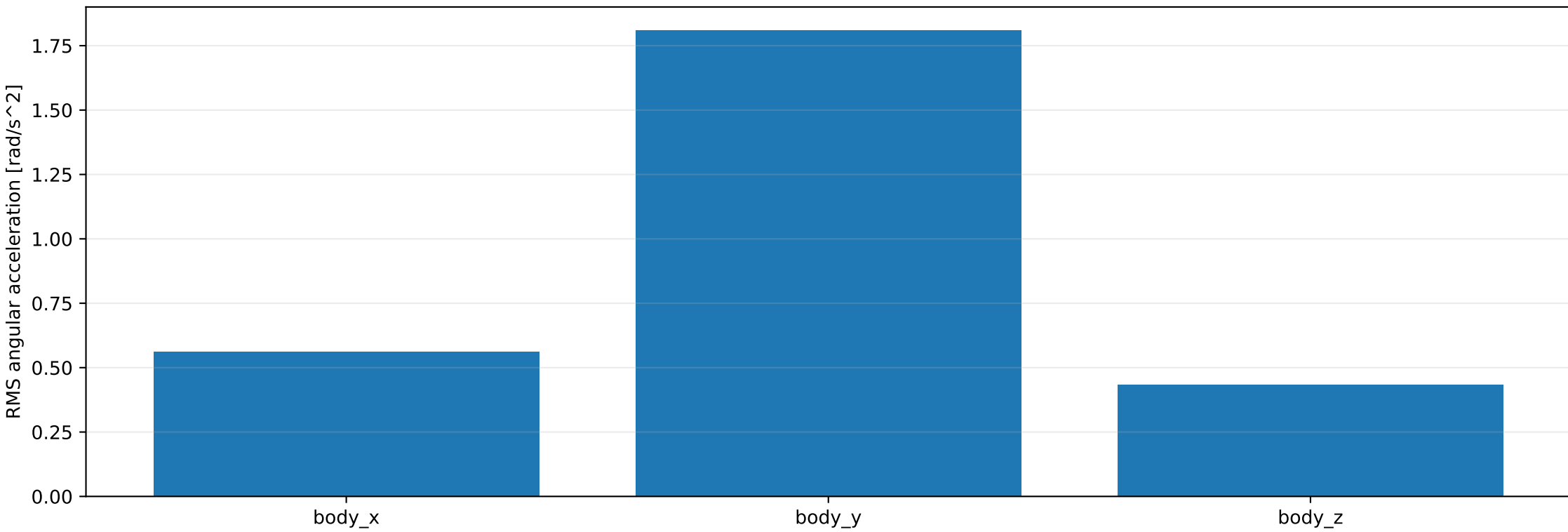
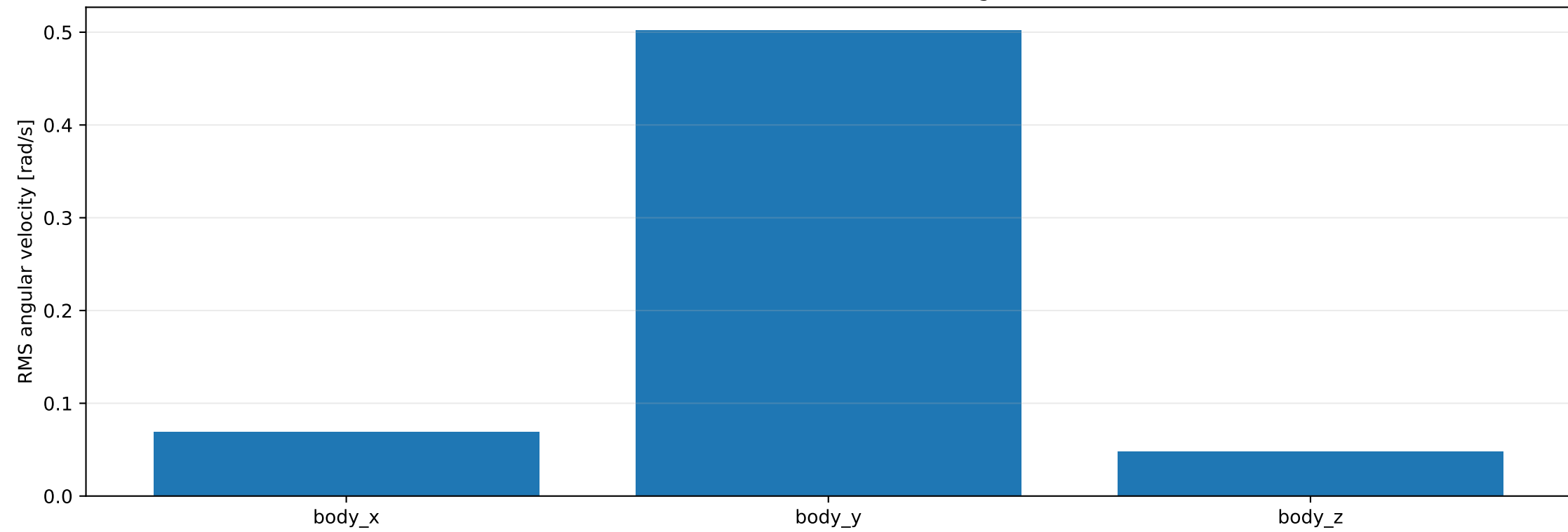
rotor force effectiveness

[0.96190153, 0.97029323, 0.96469772, 0.95671106]

selected delay [s]

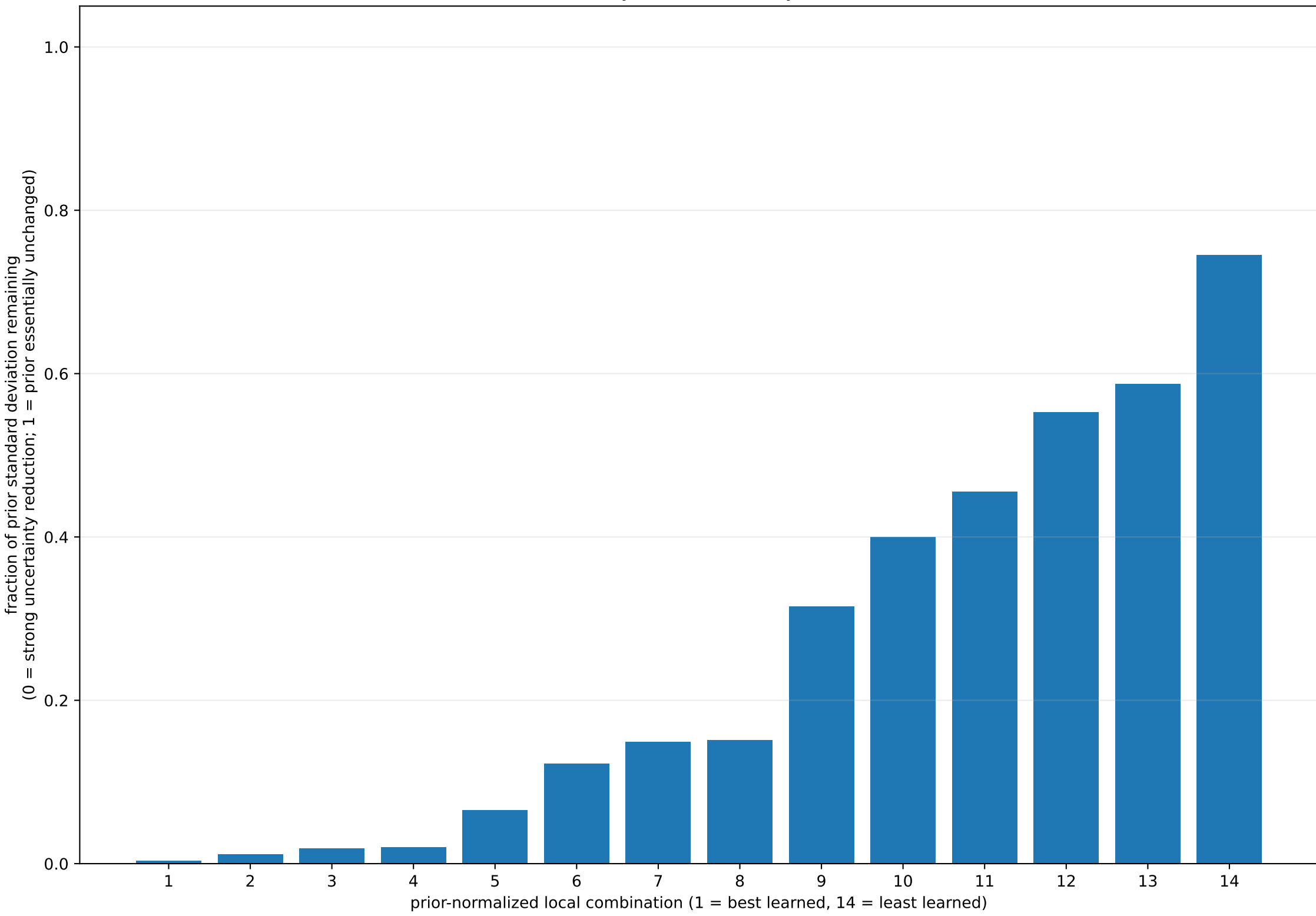
0.028301254031

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

